

Gaming PCs vs Laptops: Are laptops finally better?

Backtrack a few years ago, and I'd tell you that a gaming PC is better than a gaming laptop, hands down. It was a no-brainer; building a PC was cheaper and came with plenty of other benefits, save the portability. But is that still the case now? Gaming laptops have evolved a lot over the past few years, especially with the new RTX 40 series GPUs pushing the boundaries of what gaming laptops are capable of. So let's revisit this argument.

Now, I will say that at the highest end, gaming PCs are still better than gaming laptops. The top-of-the-line GPUs and CPUs on desktop are far superior to what you can get on a gaming laptop. That said, the most powerful gaming laptops will absolutely shred through any and all of your gaming needs anyway. Unless you were gaming at incredibly high resolutions, you wouldn't even be able to tell the difference between the two. However, it's a very small percentage of people who actually own or go for top-of-the-line specs. The majority sit at entry to mid range of gaming hardware. We're talking xx50 and xx60 series GPUs from NVIDIA, specifically, the new RTX 4060 GPU.

The RTX 4060 is unique in that both the desktop and laptop version of the GPU have the same die size. Meaning, both the laptop and desktop version have the same number of CUDA, RT, and Tensor cores. Both GPUs also have 8GB of GDDR6 VRAM, and the same 128-bit memory bus. Laptop and desktop GPUs don't usually have the same die size. As such, the only things slightly in favour of desktop GPUs is the higher memory speed and bandwidth. The desktop GPU is also more recent; it came out 4 months after the laptop GPU. Both GPUs also have the

same maximum TGP at 115W. However, while the desktop GPU is always at 115W, laptop GPUs can range from 35W to 115W.

Of course, it's more than the GPU that contributes to gaming performance. You also have the processor and RAM. With DDR5 RAM we can safely ignore RAM as a differentiating factor; as long as the capacity is the same, you should be getting identical performance. Then we come to the processor. K variant Intel processors are definitely faster than their H and HX counterparts. We won't even consider U and P variants here. This mostly has to do with desktop processors (K processors) getting more power, and therefore having better processing speeds. However, this is only really noticeable in CPU intensive titles. When GPU loads are concerned, the difference is surprisingly negligible. We're talking a mere ~5-10 FPS lower on average with RTX 4060 laptops as compared to gaming desktops. With maximum average FPS already being well over 60 in most cases, this 5-10 FPS is hardly noticeable.

That brings us to the main reason anyone considers going for either a laptop or a desktop. A laptop is a compact and portable device. Even a gaming laptop, which is far bulkier than mainstream laptops, is extremely portable when compared to a gaming desktop. With portability being the main advantage, there are disadvantages as well, for instance, laptops tend to run a lot hotter, and louder than gaming desktops. This makes sense if you consider they're outputting similar performance. The fans need to run faster (and louder) to keep the heat in check. More heat means more likelihood of throttling as well. Although to be fair, laptop cooling



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technology has also improved a lot over the years.

For the desktop, well, it's the upgradeability. Being able to upgrade components as when you need will save money in the long run. Plus, you do get slightly better FPS, and at the higher end you will definitely get better performance. But, if you were to compare prices right now for a PC build and a gaming laptop featuring the same processor, RAM and GPU, you'd find that laptops are very much competitively priced. Especially if you include the cost of a keyboard, mouse and monitor with a gaming PC.

So where does that leave us? It looks like it's fairly close. In terms of performance, budget gaming laptops are in a great place, and even comparable to gaming desktops. You can actually decide based on your need for portability or the upgradeability that a desktop offers. *